

Ahmed Khalid Humoudi

Robotics and Automation Engineer

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🌐 www.linkedin.com/in/ahmed-humoudi-904127317 🔗 portfolio-omega-ten-g6ozj4wxdx.vercel.app

PROFILE

Dedicated robotics engineering student with hands-on experience in control, automation, AI integration, and system development. Passionate and seeking opportunities to apply technical expertise to real-world applications. **Portfolio:** portfolio-omega-ten-g6ozj4wxdx.vercel.app — contains detailed project information, visualization, and implementation.

EDUCATION

German International University (GIU), Bachelor Degree, 2026 09/2022 – Present | Cairo, Egypt

Major: Robotics and Automation

GPA: 0.87 (German System), equivalent to **3.85** on the United States 4.0 GPA scale.

Ranked among the top 3 students in the Robotics Major.

Global International School (Al-Kon), High School Degree, 2022 09/2019 – 06/2022 | Jeddah, Saudi Arabia

IGCSE Score: 120%

PROFESSIONAL EXPERIENCE

Mechatronics Engineer Intern, Hiryo 02/2026 – 07/2026 | Cairo, Egypt

Designed and developed a Bluetooth-based alarm and tracking device with real-time alert and location features, integrating embedded hardware, firmware, and mobile connectivity to deliver a reliable end-to-end system.

Physics Lab Assistant, German International University 02/2024 – 06/2024 | Cairo, Egypt

Conducted hands-on laboratory supervision and technical guidance, assisting in experimental procedures and the practical application of theoretical concepts.

PROJECTS

Bachelor Thesis

Reinforcement Learning-Based Path Planning for Warehouse Robots in Dynamic Environments:

Developed an RL-based navigation system for autonomous warehouse robots operating in dynamic environments with obstacle avoidance and adaptive path planning.

Multi-Robot Task Formation: Centralized ROS2 system, obstacle avoidance, path planning, formation control.

PLC-Based Stepper Motor Motion Control: controlling stepper motor speed, direction, position control.

UR5E Pick and Place: UR5e robot integrated with ROS 2, MoveIt 2, and RViz for trajectory planning.

Oven Control System: ATmega32, temperature monitoring, fan control, and safety lock system.

Ball and Beam Control System: PID-based ball stabilization on a beam using Arduino Uno.

4-DOF Robotic Arm: Arduino-based 4-DOF arm with inverse kinematics and trajectory planning.

TECHNICAL SKILLS

Programming: PLC, Embedded C, C++, JavaScript, Python, MATLAB, Simulink, ROS, React Native

Design/Simulation: AutoCAD, SolidWorks, Fusion, Proteus, Simul-ide

Electronics: Arduino, Raspberry Pi, Atmega32, ESP32

Artificial Intelligence: Machine Learning, Deep Learning, Reinforcement Learning

Microsoft Office: Word, Excel, PowerPoint

Languages: Arabic (Native), English (C1), German (A2)

COURSES

Modern Robotics Foundations of Robot Motion, Northwestern University, Coursera Online

Introduction to Artificial Intelligence, IBM, Coursera Online

Matlab, Simulink, Reinforcement Learning Onramps, MATLAB Self-Paced Online Courses